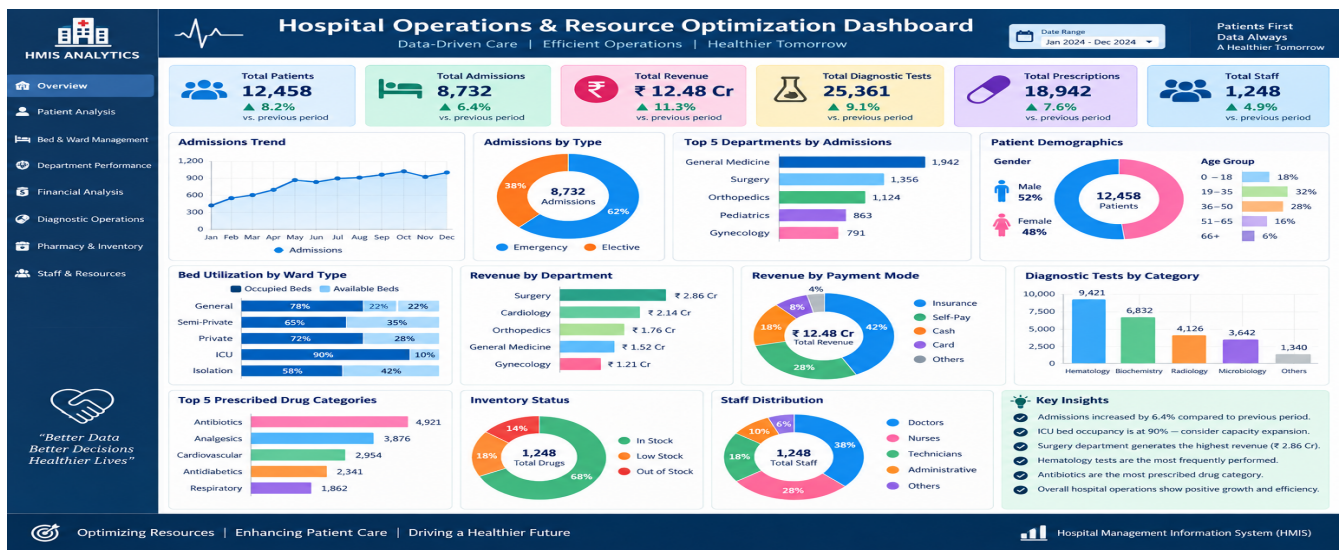


HOSPITAL HMIS MANAGEMENT INFORMATION SYSTEM ANALYSIS

SQL PROJECT • HOSPITAL OPERATIONS, FINANCIAL PERFORMANCE & RESOURCE UTILIZATION



Repository-ready documentation

An end-to-end SQL analytics project that transforms hospital management data into actionable insights using KPIs, aggregations, CTEs, CASE expressions, joins, subqueries and multi-dimensional analysis.

Designed to help a reviewer understand the hospital business problem, relational database structure, SQL workflow, analysis phases, findings and how the project can be extended.

1. Executive Summary

This project analyzes hospital operations from a Hospital Management / Administrator perspective. The database contains interconnected patient, admission, bed, ward, department, doctor, employee, billing, diagnostic, prescription, drug, inventory, insurance and manufacturer tables. The SQL work is organized into four analytical layers: KPI analysis, univariate analysis, bivariate analysis and multivariate analysis.

The purpose

The purpose is not only to calculate totals, but to explain patient demand, bed utilization, department performance, financial performance, diagnostic activity and inventory pressure so that hospital administrators can make operational decisions.

Project Snapshot

Metric	Value	Metric	Value
Unique Patients	30,000	Admissions	45,000
Bed Capacity	415	Occupied Beds	270
Departments	11	Doctors	98
Hospital Revenue	■1,684,246,109.00	Average Bill	■37,427.69
Diagnostic Tests	63,269	Prescriptions	73,109
Bed Occupancy	65.1%	Low Stock Drugs	17.6%
Data Period	01 Jan 2020 – 31 Dec 2025	Abnormal Tests	50.3%

Headline findings: Surgery is the highest-revenue department at ■377,216,234.00; ICU has the highest current bed occupancy at 80.0%; Infectious is the largest disease category by admissions; and Pathology accounts for the largest diagnostic volume.

Data note

This report uses the actual Hospital HMIS relational dataset and the 60-query SQL workflow supplied for the project. The report preserves the four-phase structure of the reference project while changing the business domain, entities, metrics and findings to hospital management.

2. Business Problem & Objectives

Business Problem

A hospital needs a reliable way to understand patient demand, resource utilization, clinical workload, financial performance, diagnostic operations and medicine inventory. Raw transactional records across many tables make it difficult to see concentration, efficiency and pressure points. This project addresses that challenge by turning relational hospital data into decision-ready summaries.

Objectives

Objective	Management Question
Measure patient demand	How many patients and admissions does the hospital handle?
Evaluate bed utilization	How much of the available bed capacity is occupied?
Compare departments	Which departments handle the greatest admission and revenue load?
Assess clinical staffing	How do doctors and staff coverage relate to workload?
Evaluate financial performance	How much revenue is generated and how much remains outstanding?
Analyze diagnostics	What is the test volume and abnormal-result pattern?
Monitor inventory	Which drug categories show higher demand relative to stock?

End Consumer: Hospital Management / Administrators

The project is framed around Hospital Operations, Financial Performance & Resource Utilization. The analysis therefore focuses on patient demand, capacity, staffing, revenue, diagnostics and inventory rather than technical database administration.

3. Database Structure

The Hospital HMIS database contains 19 interconnected tables. The model combines master/reference tables with transactional tables so that hospital activity can be analyzed across patients, admissions, resources, finance and clinical services.

Table	Role in Analysis
patient	Patient demographics and identity
admission	Patient admission, department, ward, bed and disease linkage
department	Department master and department type
ward	Ward type and total bed capacity
bed	Individual bed status and ward assignment
doctor	Doctor specialization and experience
employee	Hospital workforce and role information
staff_assignment	Employee-to-ward shift assignment
billing	Admission-level financial transactions
billing_detail	Detailed room, drug, test and procedure charges
diagnostic_test	Diagnostic test master and standard cost
patient_diagnostic	Patient test activity and result status
prescription	Admission-level medicine prescriptions
drug	Drug master, category and unit cost
drug_inventory	Current stock, reorder level and inventory status
disease	Disease and disease category
patient_insurance	Patient policy and coverage information
insurance_provider	Insurance provider master
drug_manufacturer	Drug manufacturer and reliability information

Data Characteristics

The supplied dataset contains 30,000 patients, 45,000 admissions, 415 beds, 11 active departments, 98 doctors, 63,269 diagnostic records and 73,109 prescriptions. The admission period spans 01 Jan 2020 to 31 Dec 2025.

Logical Model

Business Area	Used by SQL for
Patient	Gender, blood group, demand and admission patterns
Admission	Volume, type, length of stay and departmental workload
Beds / Wards	Capacity, occupancy and staff-to-bed analysis
Department / Doctor	Clinical workload, doctor coverage and revenue
Billing	Revenue, bill value, payment mode and outstanding amount
Diagnostics	Test volume and abnormal-result analysis
Drugs / Inventory	Prescription demand, stock and category pressure

4. SQL Methodology

The query file is deliberately staged from simple hospital KPIs to deeper multi-dimensional analysis. This makes the project easy to explain during a portfolio review or interview.

Technique	How it is used
SELECT / WHERE / ORDER BY	Retrieve, filter and rank hospital result sets
JOINS	Connect patients, admissions, wards, departments, billing and clinical tables
GROUP BY + Aggregations	SUM, AVG, COUNT, MIN and MAX for operational summaries
CTEs (WITH)	Break complex occupancy, revenue and ratio calculations into readable steps
Subqueries	Compare grouped metrics with maxima, totals and reference values
CASE	Create occupancy, payment, stock and length-of-stay categories
Window Functions	Rank entities and compare periods within groups
Date Functions	Analyze admission timing and length of stay
Derived Metrics	Occupancy %, revenue per admission, tests per admission and demand-to-stock ratio

Analysis Pipeline

Raw hospital data → Relational database → KPI layer → Univariate distributions → Bivariate relationships → Multivariate patterns → Business insights

Phase 1 establishes the baseline; Phase 2 describes individual dimensions; Phase 3 studies relationships between two dimensions; and Phase 4 combines three or more dimensions to reveal stronger operational patterns.

5. Phase 1 — KPI Analysis

Phase 1 establishes the executive baseline using 15 KPI queries. These queries cover patient volume, admissions, capacity, staffing, revenue, diagnostics, inventory and prescription activity.

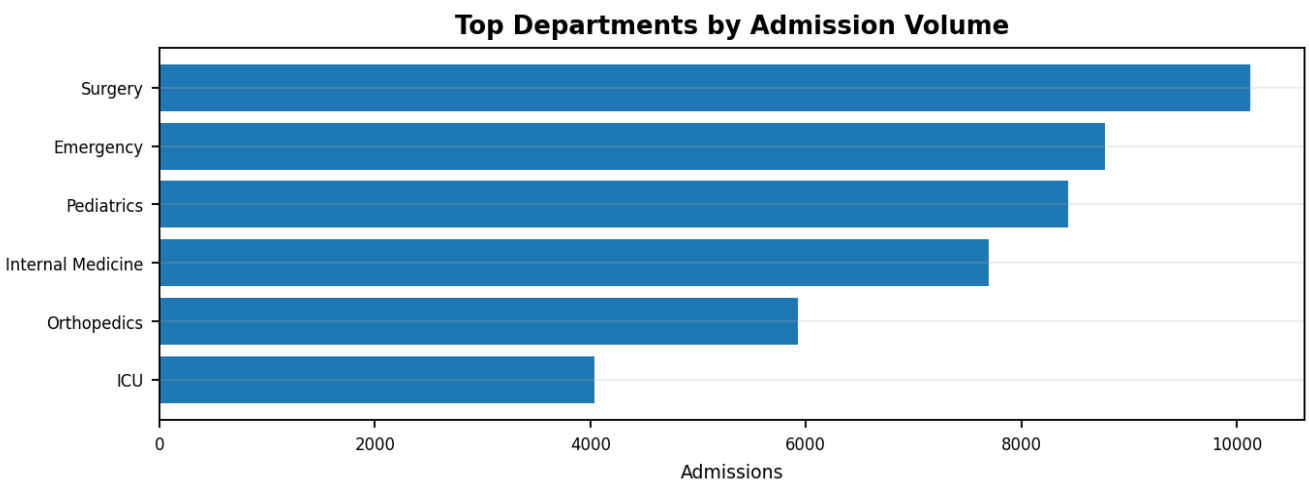
KPI	Result	Why it matters
Total Unique Patients	30,000	Patient base
Total Admissions	45,000	Hospital activity
Average Admissions / Patient	1.50	Repeat utilization
Total Bed Capacity	415	Available capacity
Bed Occupancy Rate	65.1%	Capacity utilization
Active Departments	11	Service coverage
Average Admissions / Department	4090.91	Department workload
Doctor-to-Admission Ratio	459.18	Clinical workload
Total Hospital Revenue	■1,684,246,109.00	Financial output
Average Bill Value	■37,427.69	Typical billing size
Outstanding Bill Amount	■356,395,247.60	Receivables exposure
Total Diagnostic Tests	63,269	Diagnostic workload
Abnormal Diagnostic Rate	50.3%	Clinical signal
Low Stock Drug Rate	17.6%	Inventory risk
Total Prescription Volume	73,109	Medicine demand

Interpretation

The KPI layer is intentionally simple. It provides the figures that hospital management would place on an executive dashboard before drilling into the operational reasons behind performance.

6. Phase 2 — Univariate Analysis

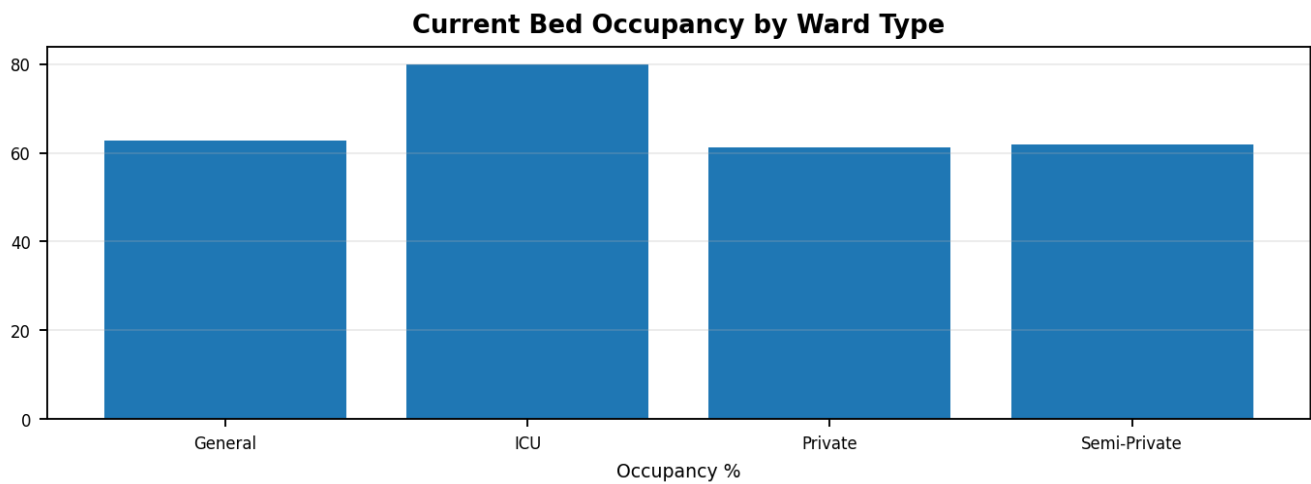
Phase 2 analyzes one hospital dimension at a time. The query file contains 15 analyses covering patient demographics, admissions, beds, wards, departments, workforce, billing, diagnostics, drugs and inventory.



Rank	Department	Admissions
1	Surgery	10,126
2	Emergency	8,777
3	Pediatrics	8,438
4	Internal Medicine	7,695
5	Orthopedics	5,924
6	ICU	4,040

Department & Bed Utilization

The univariate results highlight how hospital activity is distributed across departments and current bed states.

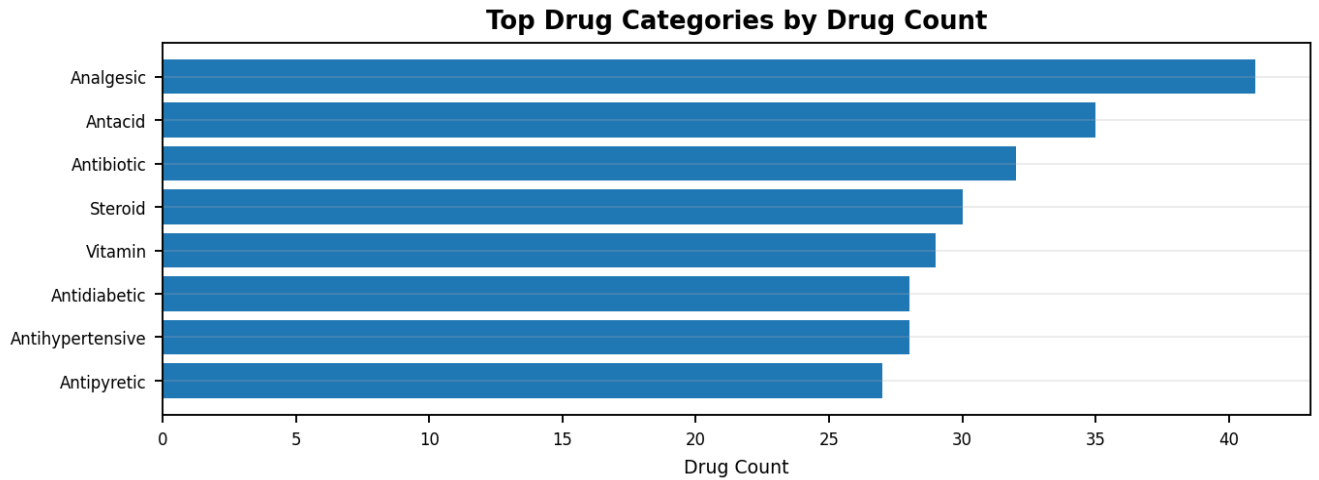


Ward Type	Total Beds	Occupied Beds	Occupancy
ICU	65	52	80.0%
General	220	138	62.7%
Semi-Private	55	34	61.8%
Private	75	46	61.3%

Highest occupancy: ICU at 80.0%.

Drug & Inventory Distribution

Inventory-focused univariate analysis describes the breadth of the drug portfolio and the current split between normal and low-stock items.



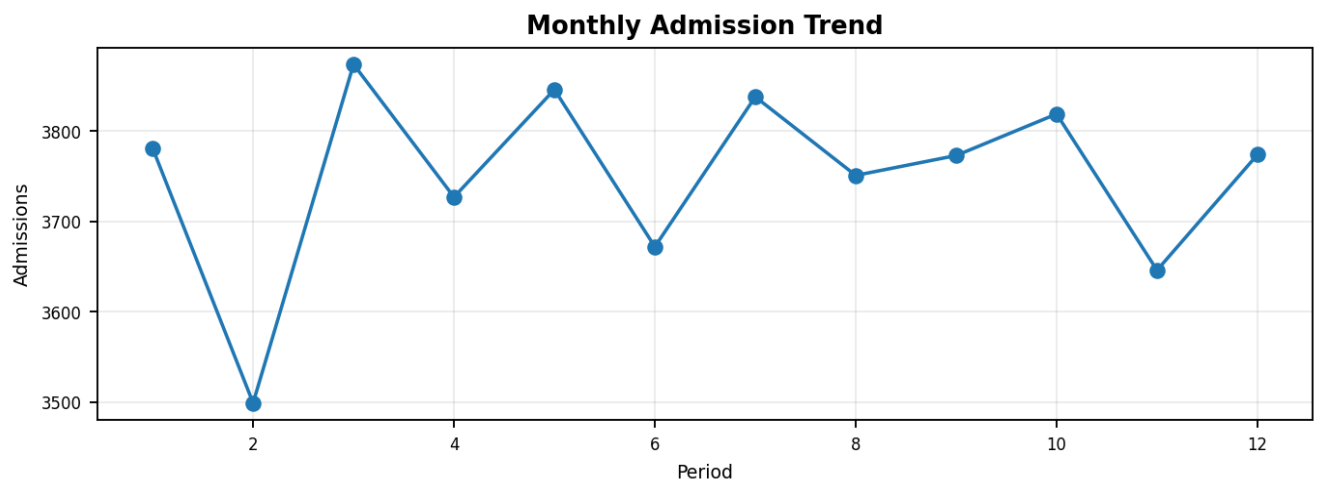
Drug Category	Drug Count
Analgesic	41
Antacid	35
Antibiotic	32
Steroid	30
Vitamin	29
Antihypertensive	28
Antidiabetic	28
Antipyretic	27

Inventory Status

Inventory Status	Drug Count
Normal	206
Low	44

7. Time Analysis — Phase 2

The admission dataset spans 01 Jan 2020 to 31 Dec 2025. Monthly and quarterly comparisons identify changes in patient demand during the available period.

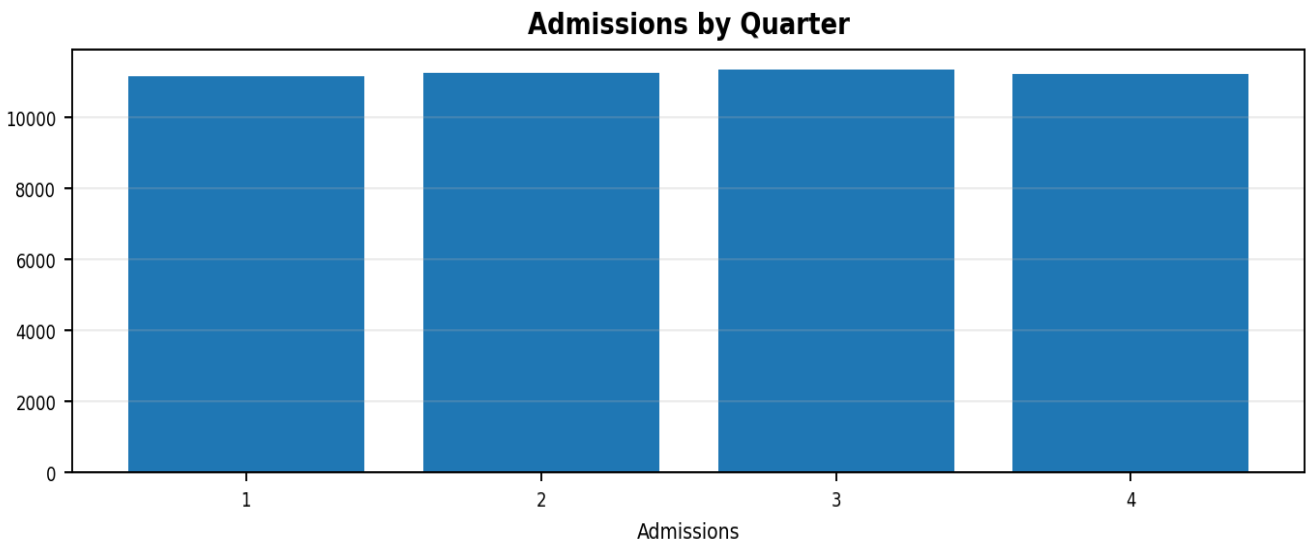


Month	Admissions	Share
1	3,781	8.4%
2	3,499	7.8%
3	3,874	8.6%
4	3,727	8.3%
5	3,846	8.5%
6	3,672	8.2%
7	3,838	8.5%
8	3,751	8.3%
9	3,773	8.4%
10	3,819	8.5%
11	3,646	8.1%
12	3,774	8.4%

Strongest month: Month 3 with 3,874 admissions. Weakest month: Month 2 with 3,499 admissions.

Quarterly Admission Trend

Quarter-level aggregation provides a compact view of demand across the available reporting period.



Quarter	Admissions	Share
Q1	11,154	24.8%
Q2	11,245	25.0%
Q3	11,362	25.2%
Q4	11,239	25.0%

Interpretation

Q3 is the strongest quarter with 11,362 admissions, while Q1 is the lowest within the available data window. These comparisons help administrators align staffing and capacity planning with demand patterns.

8. Phase 3 — Bivariate Analysis

Phase 3 studies relationships between two dimensions. Instead of asking only 'how many?', the queries ask how patient demand, capacity, staffing, finance, diagnostics and inventory behave across paired hospital dimensions.

Analysis	Business interpretation
Admission Type vs Department	Compare elective and emergency demand across departments.
Gender vs Admission Type	Compare patient gender mix across admission types.
Disease Category vs Admission Volume	Identify disease categories driving hospital admissions.
Ward Type vs Bed Occupancy	Compare current occupancy across ward types.
Department vs Bed Capacity	Measure capacity assigned to each department.
Ward Type vs Staff Coverage	Compare assigned staff across ward types.
Ward Type vs Staff-to-Bed Ratio	Assess staffing intensity relative to beds.
Department vs Doctor Count	Compare physician coverage across departments.
Specialization vs Diagnostic Volume	Identify specialties associated with test activity.
Department vs Revenue per Admission	Compare financial efficiency across departments.
Payment Mode vs Revenue	Assess revenue mix by payment channel.
Charge Type vs Total Charge	Identify major components of hospital charges.
Payment Status vs Outstanding Amount	Locate receivables exposure.
Test Category vs Abnormal Result Rate	Compare abnormal-result patterns across test categories.
Drug Category vs Demand-to-Stock Ratio	Identify inventory pressure by drug category.

Example Finding: Ward Type vs Bed Occupancy

Ward Type	Total Beds	Occupied Beds	Occupancy
ICU	65	52	80.0%
General	220	138	62.7%
Semi-Private	55	34	61.8%
Private	75	46	61.3%

This type of analysis helps management identify which ward categories are under greater capacity pressure rather than assuming all beds are utilized equally.

9. Phase 4 — Multivariate Analysis

Phase 4 is the strongest analytical layer. It combines three or more variables and uses CTEs, joins, CASE logic and window functions to reveal operational patterns across departments, wards, staff, finance, diagnostics and inventory.

Analysis	Variables combined	Technique
Department + Admission + Disease	Department + Admission Type + Disease Category	GROUP BY
Gender + Admission + Department	Gender + Admission Type + Department	GROUP BY
Disease + Admission + Ward	Disease Category + Admission Type + Ward Type	GROUP BY
Department + Ward Bed Utilization	Department + Ward + Beds	CTE + CASE
Department + Ward + Shift Staffing	Department + Ward + Shift + Staff	GROUP BY
Department Workload	Department + Doctors + Admissions	CTE + ratio
Department + Test Category	Department + Test Category + Results	GROUP BY
Department + Admission Financials	Department + Admission Type + Billing	GROUP BY
Payment + Status + Department	Department + Payment Mode + Status + Revenue	GROUP BY
Charge + Department	Department + Charge Type + Charges	GROUP BY
Test + Department + Result	Department + Test Category + Abnormal Rate	CTE + ratio
Specialization + Test Category	Specialization + Test Category + Results	GROUP BY
Drug + Inventory Demand	Drug Category + Inventory Status + Prescriptions	GROUP BY
Drug + Manufacturer + Inventory	Drug Category + Country + Inventory	GROUP BY
Department Scorecard	Department + Admissions + Revenue + Diagnostics	CTE + derived metrics

Top Department by Admission Volume

Department	Admissions	Revenue	Revenue / Admission
Surgery	10,126	■377,216,234.00	■37,252.25
Emergency	8,777	■329,738,229.00	■37,568.44
Pediatrics	8,438	■315,714,822.00	■37,415.84
Internal Medicine	7,695	■289,566,059.00	■37,630.42
Orthopedics	5,924	■222,585,716.00	■37,573.55
ICU	4,040	■149,425,049.00	■36,986.40

10. Key Business Insights

- **1. Patient demand is substantial.** The hospital records 45,000 admissions across 30,000 unique patients.
- **2. Bed capacity is actively utilized.** Current occupancy is 65.1% overall, with 270 of 415 beds occupied.
- **3. ICU shows the highest ward-level pressure.** Its current occupancy is 80.0%.
- **4. Department workload is uneven.** Surgery has the highest admission volume at 10,126.
- **5. Revenue concentration follows clinical workload.** Surgery is the highest-revenue department at █377,216,234.00.
- **6. Billing exposure requires attention.** Outstanding patient-payable amount is █356,395,247.60.
- **7. Diagnostic activity is high.** The dataset contains 63,269 diagnostic records, with an abnormal-result rate of 50.3%.
- **8. Inventory risk is visible.** 44 of 250 inventory records are currently classified as low stock.
- **9. Demand-to-stock analysis adds a planning layer.** Drug categories can be prioritized using prescription volume relative to current stock.
- **10. Multivariate analysis enables operational drill-down.** Combining department, admission, billing, diagnostic and resource dimensions supports targeted management action.

Management Actions Suggested

Action	Reason
Review high-occupancy wards	Plan bed availability, discharge flow and capacity buffers.
Align staffing with workload	Use admission volume and staff-to-bed ratios to identify pressure points.
Monitor outstanding bills	Track payment status and receivables by department and payment mode.
Prioritize high-demand drug categories	Use demand-to-stock ratios to support replenishment decisions.
Review diagnostic abnormality patterns	Use test-category and department analysis to support clinical operations.

11. Technical Skills Demonstrated

Skill	Demonstrated through
SQL Server	Database creation, table creation, constraints, indexes and SQL-compatible loading
Relational JOINS	Connecting 19 hospital tables for integrated analysis
Data aggregation	SUM, AVG, COUNT, MIN, MAX and GROUP BY
CTEs	Readable multi-step occupancy, revenue and ratio calculations
Subqueries	Grouped comparisons and business-rule checks
Window functions	Ranking and within-group comparisons
Conditional logic	CASE-based categories and operational flags
Date analysis	Admission dates and length-of-stay calculations
Business metrics	Occupancy %, revenue per admission, abnormal rate and demand-to-stock ratio
Analytical storytelling	Moving from KPIs to relationships and multivariate insights

How to Reproduce the Project

Step 1: Run Hospital_HMIS_Database.sql in Microsoft SQL Server to create the Hospital_HMIS database and tables.

Step 2: Load the supplied CSV files into the corresponding 19 relational tables.

Step 3: Open Queries(2).sql after the database has been created and loaded.

Step 4: Execute Phase 1 first to validate the dataset and baseline KPIs.

Step 5: Execute Phases 2–4 progressively for deeper analysis and export the result sets to CSV.

Repository suggestion

Keep the repository organized into database/, queries/, results/, dashboard/ and docs/ folders so a reviewer can reproduce the analysis quickly.

12. Query Catalog

The project contains 60 analytical queries across four phases. The catalog below gives a reviewer a quick map from query number to business purpose.

Query	Business Question / Output
P1-01	Total Unique Patients
P1-02	Total Admissions
P1-03	Average Admissions per Patient
P1-04	Total Bed Capacity
P1-05	Bed Occupancy Rate
P1-06	Total Active Departments
P1-07	Average Admissions per Department
P1-08	Doctor-to-Admission Ratio
P1-09	Total Hospital Revenue
P1-10	Average Bill Value
P1-11	Outstanding Bill Amount
P1-12	Total Diagnostic Tests
P1-13	Abnormal Diagnostic Test Rate
P1-14	Low Stock Drug Rate
P1-15	Total Prescription Volume
P2-01	Patient Gender Distribution
P2-02	Patient Blood Group Distribution
P2-03	Admission Type Distribution
P2-04	Admission Status Distribution
P2-05	Bed Status Distribution
P2-06	Ward Type Distribution
P2-07	Length of Stay Distribution
P2-08	Department Type Distribution
P2-09	Doctor Specialization Distribution
P2-10	Employee Role Distribution
P2-11	Billing Amount Distribution
P2-12	Payment Mode Distribution
P2-13	Diagnostic Result Distribution
P2-14	Drug Category Distribution
P2-15	Inventory Status Distribution

Query Catalog — Continued

Query	Business Question / Output
P3-01	Admission Type vs Department
P3-02	Gender vs Admission Type
P3-03	Disease Category vs Admission Volume
P3-04	Ward Type vs Bed Occupancy
P3-05	Department vs Bed Capacity
P3-06	Ward Type vs Staff Coverage
P3-07	Ward Type vs Staff-to-Bed Ratio
P3-08	Department vs Doctor Count
P3-09	Specialization vs Diagnostic Volume
P3-10	Department vs Revenue per Admission
P3-11	Payment Mode vs Revenue
P3-12	Charge Type vs Total Charge
P3-13	Payment Status vs Outstanding Amount
P3-14	Test Category vs Abnormal Result Rate
P3-15	Drug Category vs Demand-to-Stock Ratio
P4-01	Department + Admission + Disease
P4-02	Gender + Admission + Department
P4-03	Disease + Admission + Ward
P4-04	Department + Ward Bed Utilization
P4-05	Department + Ward + Shift Staffing
P4-06	Department Workload
P4-07	Department + Test Category
P4-08	Department + Admission Financials
P4-09	Payment + Status + Department
P4-10	Charge + Department
P4-11	Test + Department + Result
P4-12	Specialization + Test Category
P4-13	Drug + Inventory Demand
P4-14	Drug + Manufacturer + Inventory
P4-15	Department Scorecard

13. Conclusion

This project demonstrates how SQL can move from basic hospital aggregation to business-oriented analytical reasoning. The relational HMIS database provides a structured foundation, while the 60 queries progressively answer questions about patient demand, bed utilization, departmental performance, financial exposure, diagnostics, inventory and multi-dimensional operational patterns.

For a GitHub reviewer

The project can be understood in one flow: hospital data → relational database → SQL analysis → results → dashboard → management decisions.